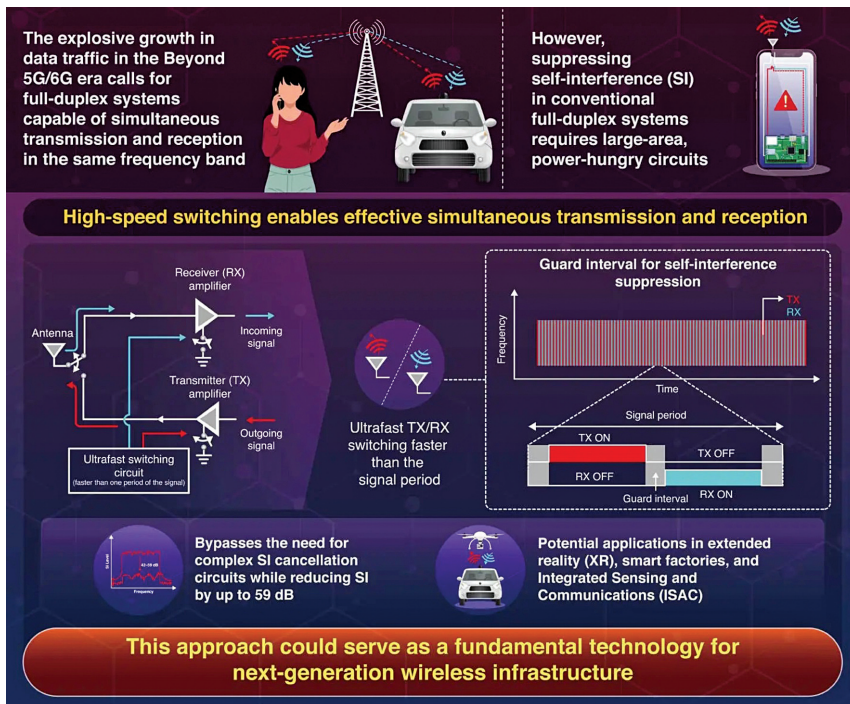


Millimetre-wave transceiver doubles spectrum efficiency

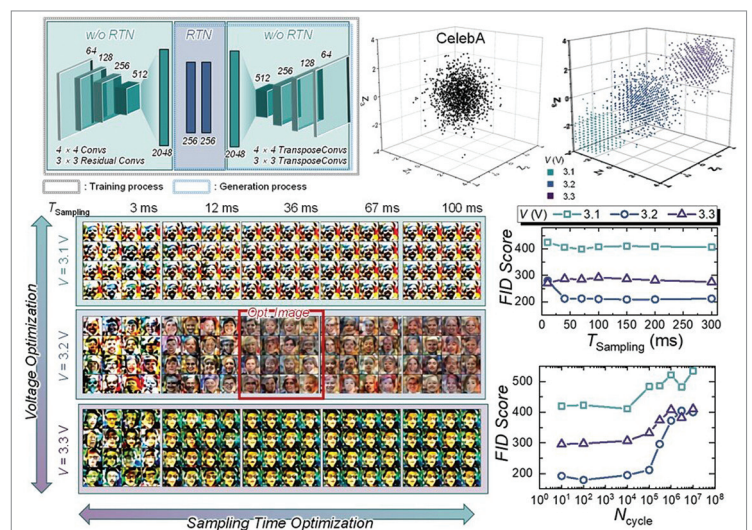


Researchers at the Institute of Science, Tokyo, have developed a millimetre-wave transceiver that enables simultaneous transmission and reception at the same frequency, potentially doubling spectrum efficiency for 5G Advanced and 6G networks. The architecture uses high-speed time-division switching to suppress self-interference without dedicated cancellation circuits, reducing hardware complexity and power consumption. Built in standard 65nm CMOS, it achieved 42dB-59dB self-interference suppression across a 400MHz bandwidth in the 24GHz-28GHz band. The technology supports AI, extended reality, autonomous systems, and smart factory applications requiring high-capacity, low-latency wireless connectivity.

Millimetre-wave transceiver (Credit: <https://www.isct.ac.jp/en>)

Sampling and deterministic computing on same chip

Researchers at Seoul National University have developed a ferroelectric memory-based semiconductor that performs both random sampling and deterministic computing on a single chip, reducing the size, power consumption, and complexity of future generative AI hardware. Developed in Seoul, the HfO₂-based ferroelectric memory switches operating modes through voltage control, enabling stochastic sampling and vector-matrix multiplication within the same memory array. The team validated the architecture using a hardware variational autoencoder that generated facial images from the CelebA dataset. The technology could enable compact, low-power edge AI processors, neuromorphic systems, and real-time on-device generative AI applications.



Noise control in ferroelectric memory and validation of image generation performance. The randomness based on RTN varies depending on voltage and sampling time conditions, which in turn affects the generated latent vector distribution and image quality (Credit: Nature Communications, 2026)